

Week 07: Learning from Data: The Machine Learning Workflow and Regression

Artificial Intelligence Applications in Engineering (MUH-920 Mühendislikte Yapay Zeka Uygulamaları). Lecture notes

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Week overview, study path and tasks: <https://github.com/utkukose/ai-applications-in-engineering-NB-lecture/blob/main/weeks/week-07/README.md>

Lecture page with animations and knowledge checks: <https://utkukose.github.io/ai-applications-in-engineering-NB-lecture/weeks/week-07/lecture.html>

Interactive lab with self-assessment and reflection: <https://utkukose.github.io/ai-applications-in-engineering-NB-lecture/weeks/week-07/lab.html>

Colab notebook with Python step 7 and the hands-on work:

https://colab.research.google.com/github/utkukose/ai-applications-in-engineering-NB-lecture/blob/main/weeks/week-07/NB07_ml_workflow_regression.ipynb

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Last update: October 2026.

Overview

The first half of the course encoded knowledge by hand: search heuristics, rules, membership functions and cost functions. Machine learning takes the opposite route and estimates a model from examples [1]. This week introduces the workflow that every data-driven engineering project follows, from the question and the data to validation and deployment [2], and applies it to regression: predicting a continuous quantity. The core case predicts the compressive strength of concrete from its mix and age, a problem studied with neural networks by Yeh [3, 4] and rooted in Abrams' water-cement rule [5]. A switch in the notebook repeats the whole pipeline on datasets from power plants, buildings, aerodynamics, gas turbines, superconductors and garment production.

Estimated study time: 9 to 11 hours.

Midterm project. The midterm project is due at the end of this week. The [project page](#) lists the deliverables and the evaluation criteria, and the report follows the common template.

Learning outcomes

By the end of the week, students are expected to describe the stages of a machine learning project, to load, inspect and clean a tabular dataset with pandas, to split data into training, validation and test sets and use cross-validation, to fit and compare linear, regularised, nearest-neighbour and tree-based regressors with scikit-learn, to report MAE, RMSE and the coefficient of determination, to read parity and residual plots, and to recognise data leakage.

Python in this week

These notes close with Python step 7: Tables with pandas and honest evaluation with scikit-learn. It covers pandas DataFrames for building, selecting, filtering, reading and summarising tables, the fit and predict pattern of scikit-learn, and error measures, baselines and cross-validation with a pipeline, applied to concrete mixes [6, 7]. In the course materials, the step is taught at the start of the Colab notebook, where each snippet runs in a cell next to its explanation and short exercises give immediate feedback, and later sections of the notebook mark further constructs as Python moves.

From rules to data

A rule-based system encodes what experts know. A machine learning system estimates a function from examples of inputs and outputs, so that it can predict outputs for new inputs [1]. In supervised learning, every example carries the correct answer: a measured strength, a failure label, a power output. In unsupervised learning, the examples carry no answers and the task is to find structure, such as groups or unusual points, which is the topic of Week 9. In reinforcement learning, an agent learns from rewards obtained by acting, which Week 14 treats. Regression is supervised learning with a continuous target, and classification, the topic of Week 8, predicts a category.

Breiman described two cultures in statistical modelling [8]. The data modelling culture assumes a stochastic model of how the data were generated and estimates its parameters; the algorithmic culture treats the mechanism as unknown and judges models by their predictive accuracy on new data. Engineering needs both views. A fitted coefficient of a physical law can be interpreted and extrapolated with care, while a random forest may predict better inside the range of the data but says little outside it.

Formulas: Learning a function from examples

Supervised learning assumes that a measured target y depends on the inputs x through an unknown function f and a random error ε with mean zero:

$$y = f(x) + \varepsilon$$

A model $\hat{f}$, read “f hat”, is chosen from a family of candidate functions so that it fits the n training pairs (x_i, y_i) well. Its average loss on the training data is the empirical risk, and for the squared loss it is the mean squared error [10]:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

The simplest model predicts the same constant c for every input. The derivative of $\sum_i (y_i - c)^2$ with respect to c is $-2 \sum_i (y_i - c)$, and it is zero when

$$c = \frac{1}{n} \sum_{i=1}^n y_i = \bar{y}$$

so the mean is the best constant under squared loss, and every other constant has a larger error by exactly $(c - \bar{y})^2$. This constant model is the baseline against which the coefficient of determination of this week measures every other model.

Worked example: The best constant for five concrete mixes

Five mixes with the water-cement ratios 0.4, 0.45, 0.5, 0.55, 0.6 reached the compressive strengths 48, 43, 40, 35, 31 MPa.

Step 1. The mean. $\bar{y} = (48 + 43 + 40 + 35 + 31) / 5 = 197 / 5 = 39.4$ MPa.

Step 2. Its error. The deviations $y_i - \bar{y}$ are 8.6, 3.6, 0.6, -4.4, -8.4, and

$$\text{MSE}(39.4) = \frac{73.96 + 12.96 + 0.36 + 19.36 + 70.56}{5} = \frac{177.2}{5} = 35.44$$

Step 3. Another constant. For $c = 40$ the deviations are 8, 3, 0, -5 and -9:

$$\text{MSE}(40) = \frac{64 + 9 + 0 + 25 + 81}{5} = \frac{179}{5} = 35.8$$

which exceeds 35.44 by $(40 - 39.4)^2 = 0.36$, as the formula predicts.

The workflow of a data-driven project

The CRISP-DM process model, developed with industrial partners, describes six phases: business understanding, data understanding, data preparation, modelling, evaluation and deployment [2]. The phases form a loop rather than a line, because evaluation often sends the team back to the data. In engineering terms, the first phase fixes the decision that the model will support and the accuracy it needs; the second checks where the data came from, which units they use and which conditions they cover; the third handles missing values, outliers and derived features; and the last phases test the model honestly and document its limits.

Data are evidence and deserve the same care as a laboratory measurement. A dataset records a particular population of mixes, machines or buildings under particular conditions. A model trained on it inherits these limits, and predictions outside the covered range are extrapolations that no validation score protects.

Check your understanding. In which phase of CRISP-DM does a team decide what decision the model will support and how accurate it must be?

- A. Business understanding
- B. Data preparation
- C. Modelling
- D. Deployment

Answer: A. The first phase translates the engineering objective into a data mining goal with success criteria [2].

Honest evaluation: Splits and cross-validation

A model that is judged on the data it was trained on looks better than it is. The standard protection is to hold out data. A test set is set aside at the start and used once, at the end, to estimate performance on new data. Model choices, such as the degree of a polynomial or the depth of a tree, are made with a validation set or with k-fold cross-validation on the training data: The training data are divided into k parts, each part serves once as validation data while the other parts train the model, and the k scores are averaged. Kohavi's study recommended ten-fold stratified cross-validation for model selection on real-world data [9].

Model complexity trades bias against variance [10, 11]. A model that is too simple misses real structure and has high error on training and new data alike. A model that is too flexible fits the noise of the training data, so its training error is low but its error on new data rises. The best complexity lies between, and only data that the model has not seen can locate it.

Animation: Underfitting and overfitting. Polynomials of growing degree are fitted to noisy samples of a smooth curve. Watch the training error fall steadily while the error on new data first falls and then rises. Run it on the [lecture page](#).

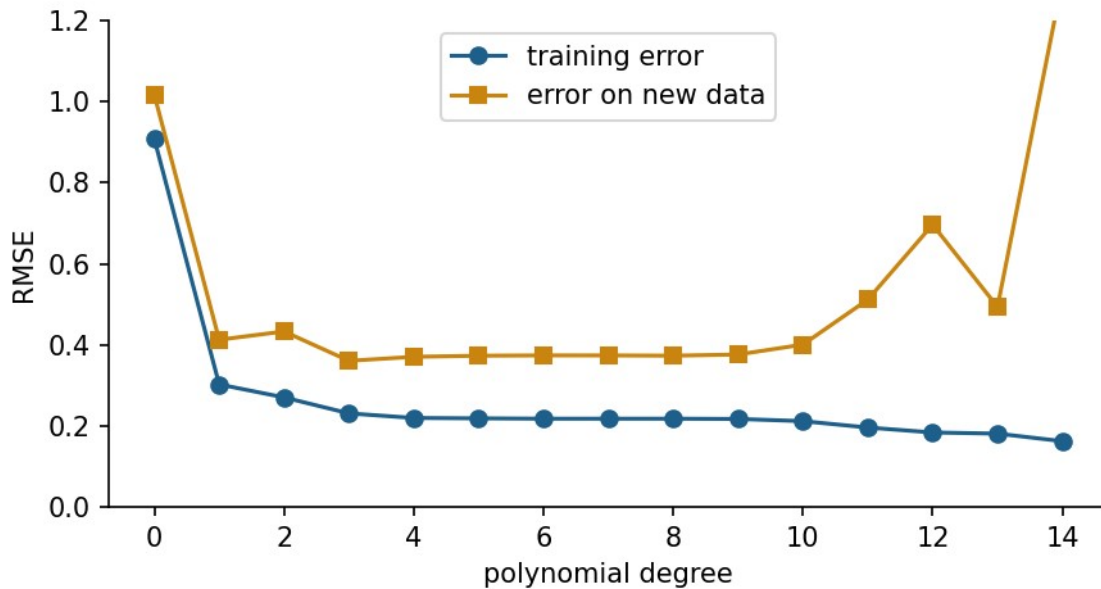


Figure 7.1. Training and test error of polynomial regression against model complexity. The gap between the curves widens as the model starts to fit noise [10].

Check your understanding. A regression tree reaches a training R-squared of 0.99 and a cross-validated R-squared of 0.61. What is the most likely diagnosis?

- A. Underfitting
- B. Overfitting
- C. Data leakage into the validation folds
- D. A perfectly calibrated model

Answer: B. A large gap between training and validation performance is the signature of variance, that is, overfitting.

Formulas: Cross-validation and the bias-variance decomposition

In k -fold cross-validation the n training examples are divided into k folds of about n/k examples each. The model is trained k times, each time on $k - 1$ folds, and its error E_j is measured on the fold j that was left out. The estimate is the mean of the k errors, and their standard deviation shows how much the estimate depends on the split [9, 11]:

$$CV_k = \frac{1}{k} \sum_{j=1}^k E_j, s = \sqrt{\frac{1}{k-1} \sum_{j=1}^k (E_j - CV_k)^2}$$

Why both too simple and too flexible models fail follows from the expected squared error at an input x_0 , averaged over the training sets that could have been drawn [10]:

$$E[(y_0 - \hat{f}(x_0))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2$$

$$\text{Bias} = E[\hat{f}(x_0)] - f(x_0), \text{Variance} = E[(\hat{f}(x_0) - E[\hat{f}(x_0)])^2]$$

The bias measures how far the average model is from the truth, the variance how much the model changes from one training set to the next, and σ^2 is the variance of the noise, which no model can remove.

Worked example: Five folds and the size of the folds

Step 1. Fold sizes. The UCI concrete data hold 1030 mixes [4]. Ten-fold cross-validation uses folds of $1030/10 = 103$ mixes and trains each model on $1030 - 103 = 927$ mixes.

Step 2. The estimate. A five-fold run gives the fold errors 4.2, 3.8, 5.1, 4.6, 3.9 MPa (MAE):

$$CV_5 = \frac{4.2 + 3.8 + 5.1 + 4.6 + 3.9}{5} = \frac{21.6}{5} = 4.32 \text{ MPa}$$

Step 3. The spread. The deviations from 4.32 are -0.12, -0.52, 0.78, 0.28, -0.42, their squares 0.0144, 0.2704, 0.6084, 0.0784, 0.1764, and

$$s = \sqrt{\frac{1.148}{4}} = \sqrt{0.287} = 0.5357 \text{ MPa}$$

A difference of 0.1 MPa between two models is therefore far inside the noise of the estimate.

Worked example: Bias and variance of two models

Suppose the true strength at one mix is $f(x_0) = 10$ MPa and the noise variance is $\sigma^2 = 1$. Four training sets give a flexible model the predictions 8, 9, 13, 10 and a rigid model the predictions 7, 7.5, 7.2, 7.3.

Step 1. The flexible model. The mean prediction is $(8 + 9 + 13 + 10)/4 = 10$, so the bias is $10 - 10 = 0$. The variance is $((8 - 10)^2 + (9 - 10)^2 + (13 - 10)^2 + (10 - 10)^2)/4 = (4 + 1 + 9 + 0)/4 = 3.5$, and the expected error is $0 + 3.5 + 1 = 4.5$.

Step 2. The rigid model. The mean prediction is $(7 + 7.5 + 7.2 + 7.3)/4 = 7.25$, so the bias is $7.25 - 10 = -2.75$ and $\text{Bias}^2 = 7.5625$. The deviations from 7.25 are -0.25, 0.25, -0.05 and 0.05, so the variance is $(0.0625 + 0.0625 + 0.0025 + 0.0025)/4 = 0.0325$, and the expected error is $7.5625 + 0.0325 + 1 = 8.595$.

The rigid model is stable but systematically wrong; the flexible model is right on average but unstable. Here the flexible model wins, while with less data its variance would grow and the comparison could turn. The animation above shows the same trade-off over the degree of a polynomial, and Part G of the [interactive lab](#) estimates both terms by repeated sampling.

Models for regression

Linear regression fits a weighted sum of the inputs by least squares. It is fast, interpretable and a mandatory baseline. Scaling the inputs does not change its predictions but matters for regularised variants. Ridge regression adds a penalty on the squared size of the coefficients, which stabilises the fit when inputs are correlated [12], and the lasso penalises absolute values, which drives some coefficients exactly to zero and selects features [13].

Nonlinear relationships need more flexible models. The k-nearest-neighbour method predicts the average target of the k most similar training examples [14]. A regression tree splits the input space into boxes with a constant prediction in each box [15]. A random forest averages many trees grown on bootstrap samples with random subsets of features, which reduces variance substantially [16]. Gradient boosting adds shallow trees one after another, each correcting the errors of the current ensemble, and implementations such as XGBoost made it a leading method for tabular data [17]. scikit-learn exposes all of these through the same `fit` and `predict` interface [7].

Formulas: Least squares, ridge, lasso, neighbours, trees and ensembles

Least squares. For one input, the line $\hat{y} = \beta_0 + \beta_1 x$ that minimises the sum of squared residuals has the slope and intercept

$$\beta_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \beta_0 = \bar{y} - \beta_1 \bar{x}$$

which follow from setting the derivatives with respect to β_0 and β_1 to zero. With many inputs collected in a matrix X , one row per example and a first column of ones, the same conditions give the normal equations and their solution $\beta = (X^T X)^{-1} X^T y$ [10].

Ridge and lasso. Ridge regression adds the penalty $\lambda \sum_j \beta_j^2$ and the lasso the penalty $\lambda \sum_j |\beta_j|$ to the sum of squared residuals, with a tuning constant $\lambda \geq 0$ [12, 13]. The ridge solution is $\beta = (X^T X + \lambda I)^{-1} X^T y$, where I is the identity matrix. For one centred input both solutions can be written down directly:

$$\beta_1^{\text{ridge}} = \frac{S_{xy}}{S_{xx} + \lambda}, \beta_1^{\text{lasso}} = \frac{\text{sign}(S_{xy}) \max(|S_{xy}| - \lambda/2, 0)}{S_{xx}}$$

Ridge shrinks the slope towards zero, while the lasso subtracts a fixed amount and sets the slope exactly to zero once $\lambda \geq 2|S_{xy}|$, which is how it selects features.

Nearest neighbours. The k -nearest-neighbour prediction is the mean target of the k training examples closest to the query x , with the Euclidean distance over the p inputs [14]:

$$\hat{y}(x) = \frac{1}{k} \sum_{i \in N_k(x)} y_i, d(x, x_i) = \sqrt{\sum_{j=1}^p (x_j - x_{ij})^2}$$

Inputs in different units must first be standardised, $z = (x - \mu)/\sigma$ with the mean μ and standard deviation σ of the training data, or the input with the largest numbers decides alone.

Trees and ensembles. A regression tree splits a node at the threshold t of an input that minimises the summed squared error of the two children, each predicting the mean of its examples [15]:

$$\text{SSE}(t) = \sum_{x_i < t} (y_i - \bar{y}_L)^2 + \sum_{x_i \geq t} (y_i - \bar{y}_R)^2$$

A random forest averages B trees, $\hat{f}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$. If every tree has the variance σ^2 and two trees are correlated with ρ , the average has the variance $\rho \sigma^2 + (1 - \rho) \sigma^2 / B$ [10, 16]. Gradient boosting starts from the mean and adds small trees h_m fitted to the current residuals, scaled by a learning rate ν (nu) [17]:

$$F_m(x) = F_{m-1}(x) + \nu h_m(x)$$

Worked example: A least-squares line through the five mixes

Step 1. Means and deviations. $\bar{x} = (0.40 + 0.45 + 0.50 + 0.55 + 0.60)/5 = 0.5$ and $\bar{y} = 39.4$. The deviations are $x_i - \bar{x} = -0.1, -0.05, 0, 0.05, 0.1$ and $y_i - \bar{y} = 8.6, 3.6, 0.6, -4.4, -8.4$.

Step 2. The two sums.

$$S_{xy} = (-0.1) \times (8.6) + (-0.05) \times (3.6) + (0) \times (0.6) + (0.05) \times (-4.4) + (0.1) \times (-8.4) = -2.1$$

$$S_{xx} = (-0.1)^2 + (-0.05)^2 + (0)^2 + (0.05)^2 + (0.1)^2 = 0.025$$

Step 3. The coefficients.

$$\beta_1 = \frac{-2.1}{0.025} = -84, \beta_0 = 39.4 - (-84) \times 0.5 = 39.4 + 42 = 81.4$$

The strength falls by 0.84 MPa for every increase of the water-cement ratio by 0.01, as Abrams' law leads one to expect [5].

Step 4. Predictions and residuals. $\hat{y} = 81.4 - 84x$ gives 47.8, 43.6, 39.4, 35.2, 31 MPa, and the residuals $y - \hat{y}$ are 0.2, -0.6, 0.6, -0.2, 0 MPa.

Step 5. Shrinkage. With $\lambda = 0.025$, ridge gives $\beta_1 = -2.1/(0.025+0.025) = -2.1/0.05 = -42$, half of the least-squares slope. The lasso with $\lambda = 2$ gives $\beta_1 = -(2.1 - 1)/0.025 = -1.1/0.025 = -44$, and with $\lambda = 4.2 = 2 \times 2.1$ the slope becomes exactly zero.

Worked example: Nearest neighbours with and without scaling

Six training mixes have a water-cement ratio, an age in days and a strength in MPa. The query mix has the ratio 0.48 and the age 28 days, and $k = 3$.

Step 1. Raw distances. For mix 1, $\sqrt{(0.40 - 0.48)^2 + (7 - 28)^2} = \sqrt{0.0064 + 441} = 21.0002$. The age dominates every distance, so the three nearest mixes are those with the same age, mixes 2, 3, 5, and the prediction is $(43 + 40 + 31)/3 = 38$ MPa.

Step 2. Standardisation. The training means are $\mu = 0.486667$ for the ratio and 45.1667 days for the age, and the standard deviations, computed with the divisor 6, are 0.071102 and 32.5598 days. Each difference to the query is divided by its standard deviation, for mix 1 $(0.40 - 0.48)/0.071102 = -1.1251$ and $(7 - 28)/32.5598 = -0.645$, so $\sqrt{-1.1251^2 + -0.645^2} = 1.2969$.

Mix	Ratio	Age (d)	Strength (MPa)	Raw distance	Scaled ratio difference	Scaled age difference	Scaled distance
1	0.4	7	35	21.0002	-1.1251	-0.645	1.2969
2	0.45	28	43	0.03	-0.4219	0	0.4219
3	0.5	28	40	0.02	0.2813	0	0.2813
4	0.55	90	41	62	0.9845	1.9042	2.1436
5	0.6	28	31	0.12	1.6877	0	1.6877
6	0.42	90	55	62	-0.8439	1.9042	2.0828

Step 3. The scaled prediction. The three nearest mixes are now 1, 2, 3, and the prediction is $(35 + 43 + 40)/3 = 39.3333$ MPa. Mix 5, whose ratio differs from the query by 0.12, left the neighbourhood, and mix 1, whose ratio differs by 0.08 and whose age differs by 21 days, entered it. After scaling, 21 days weigh as much as $0.645 \times 0.071102 = 0.046$ in the ratio. Part E of the [interactive lab](#) moves the query and shows both neighbourhoods.

Worked example: The best first split and one boosting step

Step 1. Candidate thresholds. The midpoints between neighbouring ratios are 0.425, 0.475, 0.525 and 0.575. For each, the children predict their means:

Threshold	Left strengths	Left mean	Left SSE	Right strengths	Right mean	Right SSE	Total
0.425	48	48	0	43, 40, 35, 31	37.25	84.75	84.75
0.475	48, 43	45.5	12.5	40, 35, 31	35.3333	40.6667	53.1667
0.525	48, 43, 40	43.6667	32.6667	35, 31	33	8	40.6667
0.575	48, 43, 40, 35	41.5	89	31	31	0	89

For the threshold 0.525, for example, the left SSE is

$$(48 - 43.6667)^2 + (43 - 43.6667)^2 + (40 - 43.6667)^2 = 18.7778 + 0.4444 + 13.4444 = 32.6667.$$

Step 2. The split. The threshold 0.525 gives the smallest total, 40.6667, against 177.2 without a split, so the tree first separates the two weakest mixes.

Step 3. One boosting step. Boosting starts from $F_0 = 39.4$. The residuals are 8.6, 3.6, 0.6, -4.4, -8.4, and a stump fitted to them uses the same threshold, with the means $43.6667 - 39.4 = 4.2667$ on the left and $33 - 39.4 = -6.4$ on the right. With $\nu = 0.5$:

$$F_1 = 39.4 + 0.5 \times 4.2667 = 41.5333 \text{ on the left, } F_1 = 39.4 + 0.5 \times (-6.4) = 36.2 \text{ on the right}$$

The new residuals are 6.4667, 1.4667, -1.5333, -1.2, -5.2, and the mean squared error falls from 35.44 to $(41.8182 + 2.1512 + 2.351 + 1.44 + 27.04) / 5 = 14.9601$.

Step 4. Averaging trees. With $\sigma^2 = 16 \text{ MPa}^2$ and $\rho = 0.3$, one tree has the variance 16, an average of 10 trees $0.3 \times 16 + 0.7 \times 16 / 10 = 4.8 + 1.12 = 5.92$, and an average of 100 trees $4.8 + 0.112 = 4.912$.

Averaging removes the independent part of the variance but not the correlated part, which is why random forests also decorrelate their trees with random subsets of features [16]. Part F of the [interactive lab](#) searches splits and adds boosting stumps step by step.

Measuring errors

The mean absolute error (MAE) averages the absolute differences between predictions and measurements and keeps the units of the target. The root mean squared error (RMSE) penalises large errors more strongly. The coefficient of determination R-squared compares the model with the constant prediction of the mean: A value of 1 means perfect predictions and 0 means no better than the mean. Numbers alone hide patterns, so a parity plot of predicted against measured values and residual plots against each input belong to every report. A residual trend against age, for example, reveals that the model misses the strength gain of concrete over time.

For concrete, the physics offers guidance. Abrams found that strength falls as the water-cement ratio rises [5], and strength grows with age at a decreasing rate. Yeh showed that neural networks trained on mix proportions and age predict the strength of high-performance concrete better than regression on the traditional ratio alone [3]. A derived feature such as the water-binder ratio can help simple models and makes the model easier to interpret.

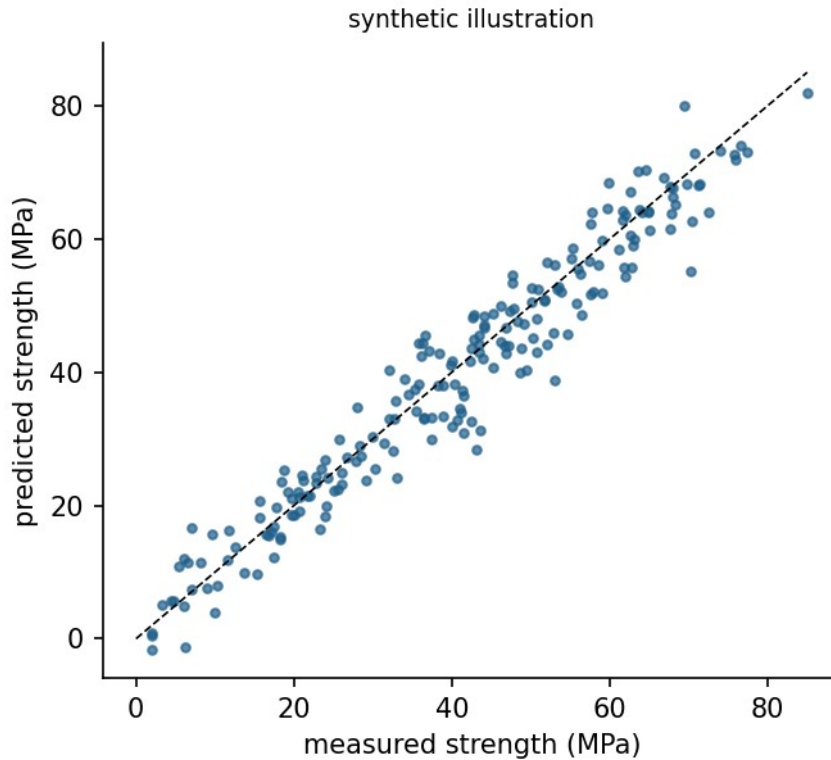


Figure 7.2. Parity plot of a gradient boosting model on held-out data of the synthetic concrete stand-in used by the notebook when the UCI data cannot be downloaded. Points on the diagonal are perfect predictions.

Check your understanding. A strength model has an MAE of 4.1 MPa and an R-squared of 0.88. Which statement is correct?

- A. On average, predictions deviate from measurements by 4.1 MPa in absolute terms
- B. 88 percent of the predictions are exactly right
- C. The model explains 4.1 percent of the variance
- D. The RMSE must also be 4.1 MPa

Answer: A. MAE is the mean absolute deviation in the units of the target; R-squared compares squared errors with those of the mean prediction.

Formulas: MAE, RMSE and the coefficient of determination

With the measured values y_i , the predictions $\hat{y}_i$ and the residuals $e_i = y_i - \hat{y}_i$ of n examples [11]:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |e_i|, \text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n e_i^2}$$

$$R^2 = 1 - \frac{\text{SS}_{\text{res}}}{\text{SS}_{\text{tot}}} = 1 - \frac{\sum_i e_i^2}{\sum_i (y_i - \bar{y})^2}$$

SS_{tot} is the squared error of the constant prediction $\bar{y}$, so R^2 is the fraction of that error which the model removes. RMSE is never smaller than MAE, and a large gap between the two signals a few large errors. Abrams' law describes the strength S of concrete through the water-cement ratio w/c with two constants A and B [5]:

$$S = \frac{A}{B^{w/c}}$$

Taking logarithms gives $\ln S = \ln A - (w/c) \ln B$, a straight line in w/c , so the constants can be fitted with least squares on $\ln S$ or, from two mixes, directly.

Worked example: The error of the least-squares line

The residuals of the line from the previous section are 0.2, -0.6, 0.6, -0.2, 0 MPa.

Step 1. MAE. $(0.2 + 0.6 + 0.6 + 0.2 + 0)/5 = 1.6/5 = 0.32$ MPa.

Step 2. RMSE. The squared residuals are 0.04, 0.36, 0.36, 0.04, 0, so $SS_{\text{res}} = 0.8$ and $RMSE = \sqrt{0.8/5} = \sqrt{0.16} = 0.4$ MPa.

Step 3. R^2 . $SS_{\text{tot}} = 177.2$ from the first example, so $R^2 = 1 - 0.8/177.2 = 0.99549$.

These are training errors on five mixes that lie almost on a line; the cross-validated errors of a real model are larger, as the knowledge check of the lecture page reminds.

Worked example: Abrams' constants from two mixes

Step 1. The constant B . Dividing $S(0.40) = 48$ by $S(0.60) = 31$ removes A : $48/31 = B^{0.60-0.40} = B^{0.2}$, so

$$\ln B = \frac{\ln(48/31)}{0.2} = \frac{\ln 1.548387}{0.2} = \frac{0.437214}{0.2} = 2.186069, B = e^{2.186069} = 8.9002$$

Step 2. The constant A . $A = 48 \times B^{0.4} = 48 \times e^{0.4 \times 2.186069} = 48 \times e^{0.874428} = 115.08$ MPa.

Step 3. A prediction. For $w/c = 0.5$: $S = 115.08/e^{0.5 \times 2.186069} = 115.08/e^{1.093034} = 38.57$ MPa, close to the 40 MPa measured. The water-binder ratio extends the idea to mixes with slag and fly ash: 180 kg of water with 300 kg of cement, 100 kg of slag and 50 kg of fly ash give $w/c = 180/300 = 0.6$ but $w/b = 180/(300 + 100 + 50) = 180/450 = 0.4$. Part D of the [interactive lab](#) computes all three metrics for any measured and predicted values.

Data leakage

Leakage occurs when information that would not be available at prediction time enters the training data [18]. Typical engineering forms are a feature computed from the target, such as a strength class assigned after testing; a measurement taken after the event to be predicted, such as the repair cost of a failure that the model should anticipate; duplicated specimens that appear in both training and test sets; and preprocessing, such as scaling, fitted on the whole dataset before splitting. Leakage produces excellent validation scores and useless models. The defence is to ask for every feature when and how it becomes known in practice, to split before any fitting, and to wrap preprocessing and model into one pipeline.

Check your understanding. A dataset for predicting pump failure contains the column “hours until repair was completed”. What is the problem?

- A. The column has the wrong units
- B. The column is known only after the failure, so using it leaks the target
- C. The column must be scaled first
- D. Tree models cannot use time columns

Answer: B. Information recorded after the event cannot be available when the prediction is needed [18].

Formulas: Fitting preprocessing without leakage

Every preprocessing step has parameters θ that are estimated from data, such as the mean and standard deviation of a standard scaler or the median used to fill missing values. Without leakage they are estimated from the training data only and then applied unchanged to new data [7, 18]:

$$\theta = g(D_{\text{train}}), z = T_{\theta}(x) \text{ for training and test data alike}$$

For the standard scaler, $\theta = (\mu, \sigma)$ and $T_{\theta}(x) = (x - \mu)/\sigma$. Estimating θ from all data lets the test examples shape the transformation that is later used to judge them.

Worked example: A scaler that has seen the test point

The training values of an input are 10, 12, 14, 16, 18, and a new test value is 30, far outside the training range.

Step 1. Correct scaling. $\mu = (10 + 12 + 14 + 16 + 18)/5 = 14$ and

$\sigma = \sqrt{(16 + 4 + 0 + 4 + 16)/5} = \sqrt{8} = 2.8284$, so the test value becomes $z = (30 - 14)/2.8284 = 5.6569$. It lies more than five standard deviations outside the training data, a clear warning of extrapolation.

Step 2. Leaky scaling. With all six values, $\mu = 100/6 = 16.6667$. The deviations are $-6.6667, -4.6667, -2.6667, -0.6667, 1.3333, 13.3333$, their squares $44.4449, 21.7781, 7.1113, 0.4445, 1.7777, 177.7769$, so $\sigma = \sqrt{253.3334/6} = 6.4979$ and the test value becomes $z = (30 - 16.6667)/6.4979 = 2.0519$.

The leaky scaler hides the extrapolation, because the test point has widened the scale that measures it. Wrapping scaler and model into one pipeline, which is fitted inside every training fold, prevents this. Part B of the [interactive lab](#) shows other forms of leakage.

Python step 7: Tables with pandas and honest evaluation with scikit-learn

Tables as DataFrames

Most engineering data arrive as tables, with one row per specimen, test or time and one column per variable. The pandas library represents such a table as a DataFrame [6]. A DataFrame can be built from a dictionary of columns, and `shape`, `head` and `describe` give a first view. The values below are invented for practice and only loosely follow the behaviour of real concrete; the notebook uses the measured Concrete Compressive Strength dataset [4].

```
import pandas as pd

mixes = pd.DataFrame({
    "cement_kg": [380, 300, 450, 320, 400, 280, 350, 420],
    "water_kg": [175, 180, 160, 190, 170, 185, 165, 150],
    "age_days": [28, 28, 28, 7, 90, 28, 7, 28],
    "strength_MPa": [44.1, 33.9, 55.2, 21.3, 52.6, 29.8, 30.4, 58.7],
})
print(mixes.shape)
print(mixes.head(3))
```

Output

```
(8, 4)
   cement_kg  water_kg  age_days  strength_MPa
0         380        175         28         44.1
1         300        180         28         33.9
2         450        160         28         55.2
```

Selecting, computing and filtering

A column is selected with its name in square brackets and returns a Series, a single labelled column. A list of names selects several columns. New columns come from arithmetic on existing ones, such as the water-cement ratio, the main factor of concrete strength. A condition on a column produces a boolean Series, and using it as an index keeps the matching rows. Conditions combine with & for and and | for or, each condition in its own parentheses.

```
mixes["wc_ratio"] = mixes["water_kg"] / mixes["cement_kg"]
at28 = mixes[mixes["age_days"] == 28]
print(at28[["wc_ratio", "strength_MPa"]].round(3))
strong = mixes[(mixes["age_days"] == 28) & (mixes["strength_MPa"] >
40)]
print(len(strong), "mixes above 40 MPa at 28 days")
print(mixes.sort_values("wc_ratio")[["cement_kg", "wc_ratio"]].head(2))
```

Output

	wc_ratio	strength_MPa
0	0.461	44.1
1	0.600	33.9
2	0.356	55.2
5	0.661	29.8
7	0.357	58.7

3 mixes above 40 MPa at 28 days

	cement_kg	wc_ratio
2	450	0.355556
7	420	0.357143

Reading files and missing values

`pd.read_csv` reads comma-separated files from a path or a web address. Empty fields become `NaN`, the marker of a missing value. `isna().sum()` counts the missing values of each column, and `dropna` removes incomplete rows. Whether removing, filling or modelling missing values is appropriate depends on why they are missing. The example reads a short text as if it were a file.

```
import io
```

```
csv_text = """cement_kg,water_kg,age_days,strength_MPa
380,175,28,44.1
300,,28,33.9
450,160,28,
"""
table = pd.read_csv(io.StringIO(csv_text))
print(table.isna().sum())
print(table.dropna())
```

Output

	cement_kg	water_kg	age_days	strength_MPa
0	380	175.0	28	44.1

dtype: int64

The scikit-learn pattern

scikit-learn gives all its models the same interface [7]. A model object is created with its settings, `fit` learns from the inputs `X` and the targets `y`, and `predict` applies the model to new inputs. `train_test_split` holds back part of the data for testing, and `random_state` makes the split repeatable. With eight invented rows, the numbers below practise the syntax only; the workflow of this week explains how models are validated properly.

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

X = mixes[["wc_ratio", "age_days"]]
y = mixes["strength_MPa"]
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.25, random_state=0)
model = LinearRegression().fit(X_train, y_train)
print(model.coef_.round(2), round(float(model.intercept_), 2))
print(model.predict(X_test).round(1), y_test.to_numpy())
```

Output

```
[-98.57  0.11] 87.01
[41.3  55.1] [30.4  55.2]
```

Measuring errors

A model is judged by its errors on data it has not seen. The mean absolute error (MAE) averages the size of the errors in the unit of the target, here MPa, and the root mean squared error (RMSE) weights large errors more strongly. Both take one line of NumPy, and `sklearn.metrics` provides the same measures. A baseline that always predicts the mean strength of the training mixes shows whether the model has learned anything at all.

```
import numpy as np
from sklearn.metrics import mean_absolute_error, mean_squared_error

pred_test = model.predict(X_test)
err = y_test.to_numpy() - pred_test
print("MAE by hand:", round(float(np.mean(np.abs(err))), 2), "MPa")
print("MAE with scikit-learn:", round(float(mean_absolute_error(y_test,
pred_test)), 2), "MPa")
print("RMSE:", round(float(np.sqrt(mean_squared_error(y_test,
pred_test))), 2), "MPa")
baseline = np.full(len(y_test), y_train.mean())
print("MAE of the mean baseline:",
round(float(mean_absolute_error(y_test, baseline)), 2), "MPa")
```

Output

```
MAE by hand: 5.5 MPa
MAE with scikit-learn: 5.5 MPa
RMSE: 7.73 MPa
MAE of the mean baseline: 12.4 MPa
```

Cross-validation and pipelines

A single split depends on which rows happen to land in the test set. K-fold cross-validation divides the data into `k` parts, trains `k` times and tests each time on a different part, which gives `k` error estimates instead of one. Preprocessing such as scaling must be learned from the training folds only; otherwise

information from the test fold leaks into the model. A pipeline chains the preprocessing and the model, so that `cross_val_score` refits both inside every fold.

```
from sklearn.model_selection import KFold, cross_val_score
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler

folds = KFold(n_splits=4, shuffle=True, random_state=0)
pipe = make_pipeline(StandardScaler(), LinearRegression())
scores = -cross_val_score(pipe, X, y, cv=folds,
scoring="neg_mean_absolute_error")
print(scores.round(2), "mean MAE:", round(float(scores.mean()), 2),
"MPa")
```

Output

```
[ 5.5   6.71  4.21 19.97] mean MAE: 9.1 MPa
```

scikit-learn treats larger scores as better, so it reports errors with a negative sign, and the minus in front turns them back into MAE values. With eight invented rows, the numbers serve to practise the pattern, which the notebook of this week applies to the full dataset.

Check your understanding. A DataFrame `df` has the columns `age` and `f`. Which expression keeps the rows with `age` 28 and `f` above 40?

- A. `df[df["age"] == 28 and df["f"] > 40]`
- B. `df[(df["age"] == 28) & (df["f"] > 40)]`
- C. `df["age" == 28]`
- D. `df.keep(age=28, f=40)`

Answer: B. pandas combines conditions element by element with `&` and `|`, and each condition needs its own parentheses. The keyword `and` cannot combine two Series.

Discipline challenges

The notebook's switch loads a regression dataset for several fields with one line. Pick the row of your department and run the whole pipeline on it, or bring your own public data.

Department	Challenge
Civil Engineering	Concrete compressive strength from mix proportions and age, UCI 165 [4].
Mechanical Engineering and Electrical and Electronics Engineering	Net electrical output of a combined cycle power plant from ambient conditions, UCI 294 [19].
Civil Engineering and Physics	Heating load of residential buildings from shape parameters, UCI 242 [20].
Automotive Engineering and Mechanical Engineering	Airfoil self-noise from frequency, angle of attack and flow speed, UCI 291 [21].
Environmental Engineering and Chemical Engineering	NOx emissions of a gas turbine from ambient and process variables, UCI 551 [22].
Physics, Chemistry and Mathematics	Critical temperature of superconductors from composition features, UCI 464 [23].
Industrial Engineering and Textile Engineering	Actual productivity of garment production teams, UCI 597 [24].
Food Engineering and Chemistry	Wine quality score from physicochemical measurements as a regression target, UCI 186 [25].

Department	Challenge
Statistics	Compare ordinary least squares, ridge and lasso paths on the concrete data and interpret shrinkage [12, 13].
Geology, Geophysics, Mining and Earth Sciences Engineering	Predict a well-log property (for example PE) from the other logs of the SEG facies data used in Week 9 [26].

Weekly task

Use the discipline switch of the notebook, or a public dataset from DATASETS.md, to build a regression model for a quantity from your department. Report the data source and its limits, the split, at least four models compared with cross-validation, the final test result, a parity plot and one residual plot, and a paragraph on possible leakage, in about 500 words. Cite at least three works from this week's references [2, 9, 18].

Research and report assignment (optional)

Machine learning for an engineering property. Review how machine learning predicts one engineering property, for example concrete strength, building energy demand, power plant output or material critical temperature. Compare the data sources, input features, models and validation schemes of at least six studies and discuss what physical knowledge the best models use [3, 19, 20, 23].

Both assignments are optional. During an active semester, the weekly task can be sent together with the learning log exported from the interactive lab, and the research report on its own, to utkukose@sdu.edu.tr or utkukose@gmail.com. They carry no separate weight, but the instructor may take them into account as a discretionary adjustment of the midterm component of the grade. The assessment section of the syllabus gives the details, and research reports follow the report template of the course repository.

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